

Explaining Variability in Oxygen Isotope Ratios in the Western North Atlantic Using a Proxy System Model

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The western North Atlantic region has recently undergone rapid ocean warming, which has already affected key species. To investigate whether this warming trend is a recent phenomenon or a long-term signal, it is necessary to look back further than modern instrumentation allows. Paleoclimate proxies like *Arctica islandica* (ocean quahog) shells offer a look into past environmental conditions via shell-derived oxygen isotopic records. However, when looking at isotopic records from six different sites in the western North Atlantic ranging from the Delmarva Peninsula to the northern Gulf of Maine, large amounts of variability exist between these regions. To investigate these differences, climate and ocean model output and ocean reanalysis products were used to generate a simulated shell oxygen isotope record, which was then compared to the measured oxygen isotope records between 1850 CE and present day. This approach, called proxy system modeling, allows certain environmental variables to be held constant so that factors that might be driving the observed oxygen isotopic differences between the sites, such as growing season, temperature, and seawater isotope variability, can be isolated. Differences between the products were also explored. For the reanalysis products, both SODA (0.5° resolution) and GLORYS (1/12° resolution) were analyzed. For model products, both a coupled climate model (iCESM; the isotope-enabled version of the global CESM1.2, with a nominal 1° horizontal resolution globally) and an ocean-only model (VIKING20X; a 1/20° Atlantic Ocean nest embedded in a 1/4° global ocean, using the NEMO3.6 configuration and forced by the JRA55-DO atmospheric product) were used. Using correlation coefficients, we compared the resulting pseudodata to the isotope records to determine primary environmental drivers of variability between each site, finding that the variability in certain sites' oxygen isotope records were primarily temperature-driven, salinity-driven, or were influenced by both temperature and salinity. The results of this study suggest that the bivalve-tailored Grossman and Ku equation was more suitable than other pseudocarbonate equations typically used for corals. In constructing these models, a better understanding of the specific environmental conditions that influence the shell oxygen isotope records from each site is gained, enabling a more robust reconstruction of past environmental changes along the western North Atlantic shelf.

1. Introduction

Over the last 100 years, the Gulf of Maine has undergone substantial changes in sea surface temperature, with the area warming more rapidly than the rest of the global oceans since 1980 (Pershing et al., 2015). These changes in temperature are causing habitat collapses and stresses on some of the most vital species in the region, including the critically endangered North Atlantic right whale (*Eubalaena glacialis*) (Record et al., 2019) and lobsters (*Homarus americanus*), whose response to this change has begun to affect the crucial lobster industry in Maine (Pershing et al., 2018). Additionally, warming in the Mid-Atlantic Bight Cold Pool has already begun to affect boreal fauna like *Arctica islandica*, which depend on the cooler temperatures of the Cold Pool to live as far south as the Mid-Atlantic Bight (Friedland et al., 2022).

In order to understand if this warming trend is something long-term or a relatively recent phenomenon, we must look back further than modern instrumentation will allow. This is where paleoclimate proxies, which record past environmental conditions, become useful tools in reconstructing past temperature patterns. Marine calcifiers, like corals and bivalves, can document seawater conditions via the oxygen isotopic signature of their carbonate shells (Williams et al., 2024). In the current study, ocean quahog (*Arctica islandica*) shells were used as a marine calcifier proxy archive.

Atoms from the same element can have a different number of neutrons, and these different atoms are called isotopes. For oxygen, the two most common types of isotopes are ^{16}O and ^{18}O , with ^{18}O being heavier than ^{16}O . These isotopes behave slightly differently depending on temperature, so when the water is colder, more ^{18}O relative to seawater is taken into the shell, and when it is warmer, more ^{16}O relative to the seawater is taken up into the shell. So, by measuring the ratio of oxygen isotopes relative to the seawater in the shell of a clam, the temperature of the water that the clam was living in can be estimated. Since the ratio of oxygen isotopes in the shell is relative to the oxygen isotope ratio of the seawater, the oxygen isotope ratio of the seawater will also affect the ratio of the shell. Therefore, the shells are not just a proxy archive for temperature, but may also reflect the oxygen isotopic composition of the seawater. Shells collected in the Gulf of Maine have been seen to have good agreement between $\delta^{18}\text{O}_{carb}$ values and temperature, suggesting that bivalve shells are a reliable temperature proxy in this region (Wanamaker et al., 2008; Schöne, 2013).

Here, we investigate previously developed oxygen isotope records from six different sites (Fig. 1) in the western North Atlantic (Walton, 2023; Sun, 2023, Whitney et al., 2022). When looking at these six clam sampling sites (Fig. 2), large amounts of variability can be seen between the oxygen isotopic records, though some similarities can be seen as well. In particular, all sites show a general warming trend in recent years, particularly the records of Jonesport, Isle au Haut, Delmarva Shelf, and Seguin. However, the record before this warming trend differs greatly based on the site. For example, at the Seguin site, a significant cooling period can be seen, while sites like Jonesport and Delmarva Shelf have been displaying a warming over the entire time.

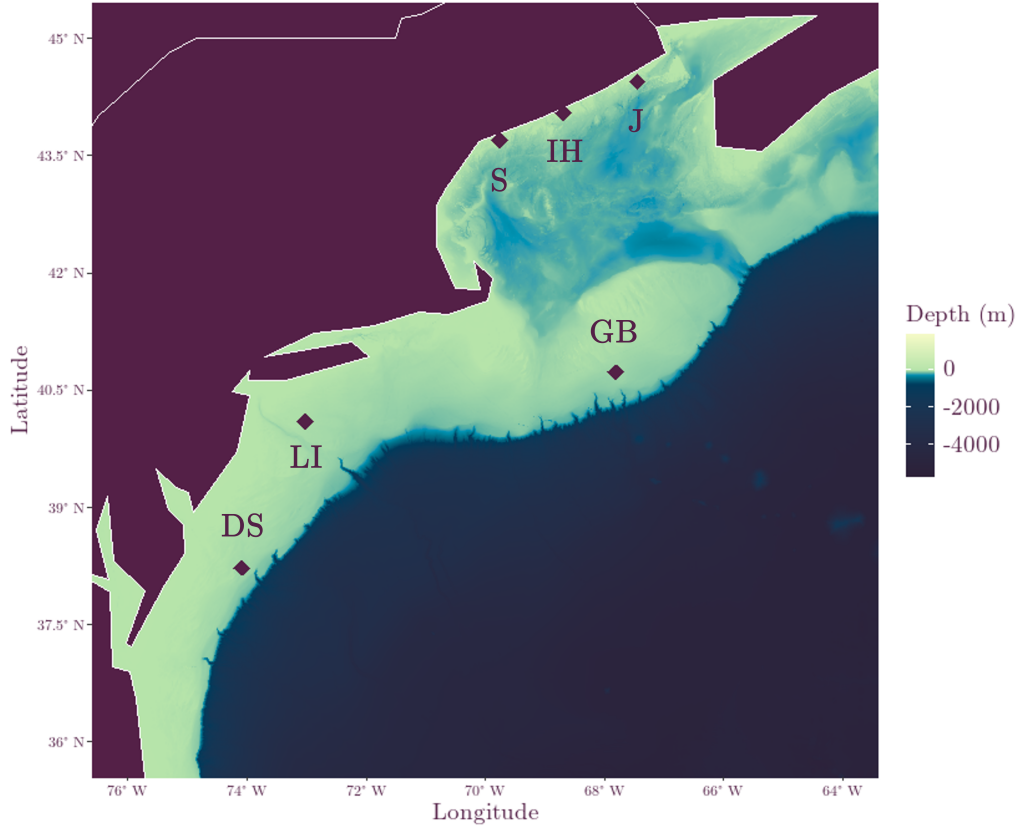


Figure 1. Selected Sites in the western North Atlantic. Site acronyms are: Jonesport (J), Isle au Haut (IH), Seguin (S), Georges Bank (GB), Long Island (LI), Delmarva Shelf (DS).

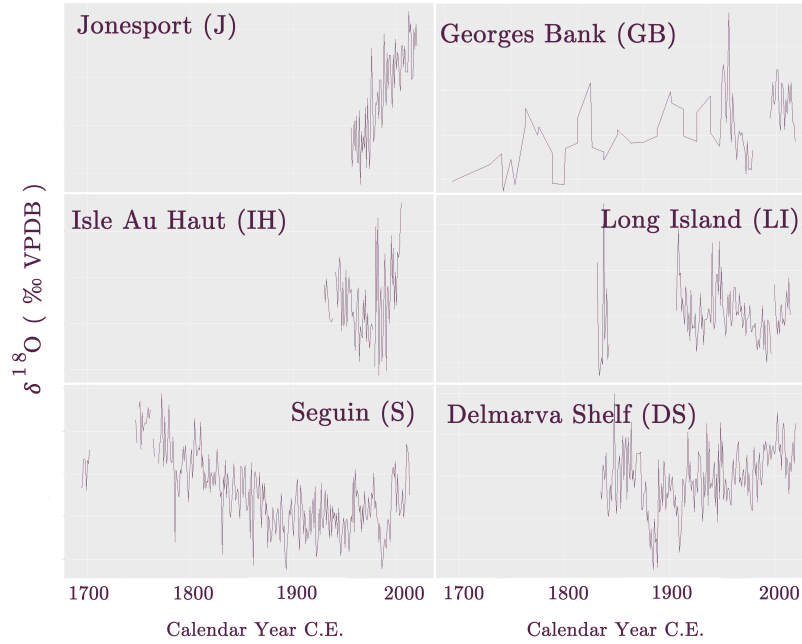


Figure 2. *Oxygen Isotope Records of Selected Sites*

This variability between sites might be due to different regions being influenced by different ocean dynamics and therefore experiencing different temperature trends. Essentially, one site's oxygen isotope record might be more influenced by seawater oxygen isotope variability while another site's oxygen isotope record might be more influenced by temperature variability. Additionally, many different factors like temperature, seawater oxygen isotope compositions, growing season of the data, or perhaps uncertainty encountered when dating the clams might be driving the differences seen between these records. To confirm what factors are causing this variability, this needs to be investigated further, and is done so in this study using a strategy called proxy system modeling.

A proxy system model, or a PSM, is built to evaluate the strength of a proxy's signal and better understand the effect of different processes on the proxy's signal (Evans et al., 2013). Proxies can be influenced by many different climate variables (Williams et al., 2024), which drives the need to investigate how these variables affect reconstructions both by themselves and in tandem with one another. Here, a bivalve proxy system model is built to help interpret the oxygen isotope data from the clam shells as a proxy for seawater temperature and oxygen isotope composition.

2. Data and Methods

2.1 Shell Isotope Records

In order to build the model, a hypothetical, simulated clam oxygen isotope record that 'lives' in the modeled ocean environment is constructed using a variety of model inputs from various different data products. Climate model output and reanalysis data containing variables like temperature and seawater oxygen isotopes allow the model to generate what the simulated clam's oxygen isotope record should look like. The modeled shell record, also called the pseudocarbonate, can then be compared to the recorded shell records, and this process can be repeated for each site. This approach allows certain variables to be held constant so that what might be driving the observed differences between sites can be isolated.

The shell isotope records used in this study come from a variety of sites in the western North

Atlantic and span a multitude of different time spans, depths, and number of shells in each record. All oxygen isotope samples were analyzed at the Stable Isotope Lab at Iowa State University using a ThermoFinnigan Delta Plus XL mass spectrometer interfaced with a GasBench II and CombiPal autosampler. Standards NBS18, NBS19 and LSVEC were used, with a long-term, average analytical uncertainty of $\pm 0.1\text{‰}$. Specific details on the development of these oxygen records can be found in multiple sources detailing the analysis of these records (Walton, 2023; Sun, 2023, Whitney et al., 2022). Brief sampling details for each site are below.

Jonesport (-67.4°W, 44.4°N)

The live adult samples used for the Jonesport records were collected in both 2009 and 2020, with shells from the former being collected at around 83 meters and the latter at around 74 meters (Walton, 2023). From the two sites, 250 samples were collected in total and four shells were used to construct the oxygen isotope record, with the cross-dated time series constructed spanning from 1956 to 2020.

Isle au Haut (-68.7°W, 44.0°N)

At Isle au Haut, seven shells collected in 2011 make up the oxygen isotope record constructed at this site. The shells were collected at a depth of 31 meters and the temporal coverage of this site spans from 1929 to 2005.

Seguin (-69.8°W, 43.7°N)

The Seguin time series was constructed from *Arctica islandica* shells collected near Seguin Island, in the years 2010, 2011, 2012, and 2014 and at a depth of 38 meters (Whitney et al., 2022). In total, 270 shells were collected, and 14 were used to construct the oxygen isotope record. The temporal coverage of the annually resolved cross-dated Seguin time series goes from 1694 to 2013, and is continuous after 1769.

Georges Bank (-67.8°W, 40.7°N)

For the Georges Bank site, the time series spans from 1756 to 2015, with pre-1950 samples being taken every five years and averaged together. The annual resolution record post-1950 is comprised of four shells collected in 2016 at a depth of 73 meters. The five year resolution record collected pre-1950 is comprised of 39 shells, with each sample taken from different shells that were collected in 2017, also at a depth of 72 meters.

Long Island (-73.0°W, 40.1°N)

At Long Island, five shells collected in 2017 make up the oxygen isotope record constructed at this site. The shells were collected at a depth of 48 meters and the time series constructed for this site spans from 1830 to 2016.

Delmarva Shelf (-74.1°W, 38.2°N)

The shells collected for the Delmarva Shelf record were sampled at a depth of 63 meters, off the coast of Ocean City, MD in 2021, with a total of 233 samples being caught and five being used to construct the oxygen isotope record (Sun, 2023). The temporal coverage at this site spans from 1835 to 2020, and five shells were used to construct the oxygen isotope record.

2.2 Data Products

In order to build each of the proxy system models, both climate and ocean model output and ocean reanalysis products were used. Each product had its own unique procedure for processing the raw data and then formatting it properly to be fed into the PSM. The reanalysis products analyzed were both SODA (Carton & Giese, 2008) and GLORYS (E.U. CMEMS, 2018). The SODA data (version 3.15.2) has a 0.5° resolution and is an ocean-only model incorporating many gridded ocean variables, starting in 1980 and going until 2019. This dataset used a model forecast that is based on the GFDL MOM6 model, and is forced by the ERA5 reanalyses. The GLORYS dataset has a $1/12^\circ$ resolution and covers the time span from 1993 onward. The model component for GLORYS is the NEMO platform, which is driven by ECMWF ERA-Interim and then in recent years, the ERA5 reanalyses. A reduced-order Kalman filter was then used to assimilate the observations.

For the model products, a coupled climate model, iCESM (Brady et al., 2019), and an ocean-only model, VIKING20X, were used. The iCESM model is the isotope-enabled version of the global Community Earth System Model, CESM1.2, with a nominal 1° degree horizontal resolution globally. The model used is the Parallel Ocean Program version 2 (POP2) ocean model component of iCESM, and enabled the use of modelled seawater isotope data in the resulting proxy system models. The iCESM provides both isotope-enabled data as well as salinity and temperature data from the CESM. The seawater isotope data taken from the iCESM in this study were selected from run 3 of the model, and the temperature and salinity were taken from run 10 of the CESM. This model spans from 1850 to 2005. VIKING20X (Biaostoch et al., 2021) is a $1/20^\circ$ resolution Atlantic Ocean nest embedded in a $1/4^\circ$ global ocean, using the NEMO3.6 configuration and forced by the JRA55-DO atmospheric product. The temporal coverage of VIKING20X spans from 1958 to 2019.

2.3 Data Processing

The SODA data files were downloaded individually on an annual basis, and years were then merged and the western North Atlantic region selected. The SODA cell nearest to each specific site was then identified, and the measurements for each site then came from the midpoint of each cell that was selected. Within the specific cells, the depth of each site’s data was also ‘selected’ by using the nearest depth available to the bottom depth of the site (Table 1).

Table 1. *Selected SODA Data Cells for Each Site*

Site	Selected Cell ($^\circ\text{E}$, $^\circ\text{N}$)	Selected Depth (m)
Jonesport	(292.75, 44.25)	76.803
Isle Au Haut	(291.25, 44.25)	35.358
Seguin	(290.25, 43.75)	35.358
Georges Bank	(292.25, 40.75)	76.803
Long Island	(287.25, 40.25)	45.576
Delmarva Shelf	(285.75, 38.25)	66.262

In a similar fashion, the GLORYS data was downloaded and sliced, but due to the increased spatial resolution of the reanalysis dataset, the cells chosen were more local to the site itself. Specifically, around each site’s coordinates, a $1/6^\circ$ box was selected with the site location at its midpoint. Then, a specific depth was once again selected on the basis of being nearest to the actual depth of the site (Table 2).

Table 2. *Selected GLORYS Data Cells for Each Site*

Site	Selected Cell ($^{\circ}\text{E}$, $^{\circ}\text{N}$)	Selected Depth (m)
Jonesport	$(-67.4 \pm 1/12, 44.4 \pm 1/12)$	77.854
Isle Au Haut	$(-68.7 \pm 1/12, 44.0 \pm 1/12)$	34.434
Seguin	$(-69.8 \pm 1/12, 43.7 \pm 1/12)$	40.344
Georges Bank	$(-67.8 \pm 1/12, 40.7 \pm 1/12)$	77.854
Long Island	$(-73.0 \pm 1/12, 40.1 \pm 1/12)$	47.374
Delmarva Shelf	$(-74.1 \pm 1/12, 38.2 \pm 1/12)$	65.807

While the iCESM and VIKING20X followed similar protocols in terms of data access, selecting the specific cells and depths required more steps due to the method in which null values are handled in each model. The iCESM is gridded on a curvilinear scale, so selecting specific cells is not as straightforward. Therefore, either the cell midpoint closest to the site was chosen, or, in the case of the two southernmost sites, the nearest two cells were chosen and averaged since these sites straddled two different cells. In order to find the bottom temperature in the model, a forward fill was used to remove the NaN values and fill them with the true bottom depth values. In VIKING20X, a similar method was used, though in that model, NaN values are expressed as zeroes. Once the forward fill was completed, the bottom depth slice was then selected and the values were taken from there. The midpoints of the iCESM cells are shown in Table 3, while the VIKING20X cell boundaries are shown in Table 4. Note that in similar fashion to the GLORYS dataset, a $1/10^{\circ}$ box was selected with the site location at its midpoint for the VIKING20X data.

Table 3. *Selected iCESM Data Cells for Each Site*

Site	Selected Cell ($^{\circ}\text{E}$, $^{\circ}\text{N}$)	Selected Depth (m)
Jonesport	$(-67.694, 44.611)$	75.0
Isle Au Haut	$(-68.822, 44.240)$	35.0
Seguin	$(-69.972, 43.653)$	35.0
Georges Bank	$(-67.994, 40.970)$	75.0
Long Island	$(-72.472, 40.059)$	45.0
Delmarva Shelf	$(-74.802, 38.372)$	65.0

Table 4. *Selected VIKING20X Data Cells for Each Site*

Site	Selected Cell ($^{\circ}\text{E}$, $^{\circ}\text{N}$)	Selected Depth (m)
Jonesport	$(-67.4 \pm 1/20, 44.4 \pm 1/20)$	77.625
Isle Au Haut	$(-68.7 \pm 1/20, 44.0 \pm 1/20)$	32.209
Seguin	$(-69.8 \pm 1/20, 43.7 \pm 1/20)$	41.482
Georges Bank	$(-67.8 \pm 1/20, 40.7 \pm 1/20)$	77.625
Long Island	$(-73.0 \pm 1/20, 40.1 \pm 1/20)$	51.945
Delmarva Shelf	$(-74.1 \pm 1/20, 38.2 \pm 1/20)$	63.879

2.4 Pseudocarbonate Equations

To investigate variability, the products were used to generate a simulated shell oxygen isotope record. This process involves plugging certain temperature, salinity, and sometimes seawater isotope values from the data products into an equation and generating a pseudocarbonate based on the output. In this study, two different equations were used (Williams et al., 2024; Grossman and Ku, 1986).

Both equations are based on the concept that the oxygen isotopes measured in the clam's carbonate shell is related to both the temperature and the isotopic composition of the seawater surrounding the clam. In the Williams et al. paper, this relationship is characterized as the following:

$$\delta^{18}O_{carb} = a_1 \cdot temperature + \delta^{18}O_{sw} \quad (1)$$

The predicted value for a_1 is around 0.22‰ (Evans et al., 2000), while the value of $\delta^{18}O_{sw}$ is typically unknown. Therefore, in order to estimate $\delta^{18}O_{seawater}$, the value is estimated to be $a_2 \cdot salinity$, and the equation below is used

$$\delta^{18}O_{carb} = a_1 \cdot temperature + a_2 \cdot salinity \quad (2)$$

where a_2 is the slope of the salinity- $\delta^{18}O_{sw}$ relationship, which varies based on the sub-basin of the target region. For the western North Atlantic, this value is estimated to be 0.55 (LeGrande & Schmidt, 2006). Note that the Williams et al. paper was not the first to establish this equation or the concept of proxy system modeling, but contains the most complete representation of the equation used in this study. From this point forward, this equation will be referred to as the 'Williams et al. equation', but that naming convention is not an attribution so much as a moniker of convenience. It is also important to note that the iCESM model contains isotope-enabled data, and the PSM resulting from this data is able to use (1) without needing to approximate the value of $\delta^{18}O_{sw}$ (although salinity data is available and was also used to generate salt-enabled PSMs). All other products, however, use (2) since only salinity data is available for each dataset.

The Grossman and Ku equation is calculated under the same assumption of the relationship between carbonate oxygen isotopes, temperature, and seawater isotopic composition, but its equation takes a different form:

$$\delta^{18}O_{carb} = [(20.6 - temperature)/4.34] + (\delta^{18}O_{sw} - 0.27) \quad (3)$$

For this study, the $\delta^{18}O_{seawater}$ term is calculated using the equation below:

$$\delta^{18}O_{seawater} = 0.5 \cdot salinity - 17.3 \quad (4)$$

which is also specific to water samples taken from the Gulf of Maine (Whitney et al., 2020) and relies on the linear relationship between oxygen isotopes of seawater and salinity as well. Similar to the Williams et al. equation, if the iCESM isotope-enabled data is used, then $\delta^{18}O_{sw}$ does not need to be calculated using (4) and can be directly substituted in for $\delta^{18}O_{sw}$.

2.5 Proxy System Modeling

When building the proxy system model, a different model was built for each product (SODA, GLO-RYS, iCESM, VIKING20X) and equation (Williams et al. and Grossman and Ku) combination for each specific site. Within each of those models, the components of the model were separated so that each model ran with only temperature as an input, only salinity or isotope data as an input, and with all model inputs present. When a model input was removed, the average metric taken over the entire time period of the product for the specific input was put in its place to neutralize its effect on the variability of the model. This resulted in a total of 360 models, with the iCESM product having twice as many models due to the extra models from the isotope-enabled data. This is due to the fact that the iCESM models could be constructed using either isotope data or salinity data, and that switching one out for the other merited an entirely new set of models to be generated.

An important distinction to make between the two equations is that while Williams et al. pseudocarbonate were calculated in anomaly space, the Grossman and Ku pseudocarbonates were not. This

was done to follow the precedent of the past literature working with each of these two equations, seeing as the Williams et al. paper works in anomaly space while the Whitney et al. paper does not. Additionally the concept of a ‘growing season’ was established in the data. Either the data was left in an annual season, or calendar year format (January-December), or modified to group years by an expert growing season (October-September). This is due to the fact that in this region it has been shown that clams drop their growth bands in the fall (Whitney et al., 2022), thus marking the ‘start’ of their year, hence the basis on which the growing season was established.

After all 360 models were built, the data from the observed records and the pseudocarbonates were compared using correlation coefficients and p-values at a significance level of $\alpha = 0.05$. The p-values were calculated to account for autocorrelation in the data using a specialized method in the Pyleoclim package (Khider et al., 2022). The root mean square error and null root mean square error were also calculated between the pseudocarbonate and the shell record. For reference, the null root mean square error was calculated by comparing the oxygen isotope shell record to a pseudocarbonate containing only zeroes to see if there was any difference from the standard RMSE with the true pseudocarbonate. All statistical measures were only calculated on the years that shell record data was available, so that both the pseudocarbonate and shell record were present at each point, though the pseudocarbonate is calculated regardless of the temporal coverage of the records.

3. Results

3.1 Overview of the Sites

Out the 360 models constructed, 49 of the models were well-modeled (Fig. 3), that is, had both a significant p-value and a positive correlation coefficient, and were examined further at the level of each site (Table 5). A model with a significant positive correlation coefficient demonstrates that the pseudocarbonate and shell record values are closely related at each year, which is the overall goal of a pseudocarbonate record. From this point forward, models that fit this criteria will be referred to as ‘well-modeled’. A complete list of model statistics for all models can be seen in the Appendix (Table 1A).

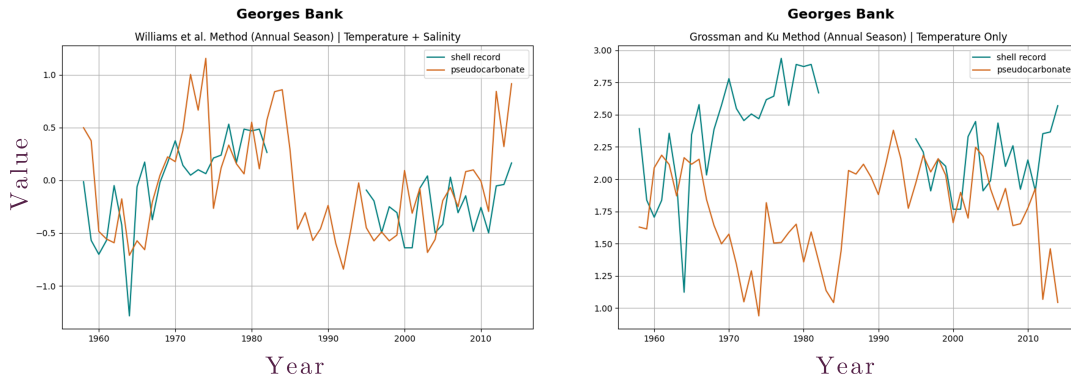


Figure 3. A well-modeled pseudocarbonate (left) and a poorly modeled pseudocarbonate (right)

Table 5. Well-Modeled Proxy System Models by Site

Site	No. Well Modeled	Percent Well-Modeled
Jonesport	15	25
Isle Au Haut	5	8.33

Site	No. Well Modeled	Percent Well-Modeled
Seguin	10	16.67
Georges Bank	4	6.67
Long Island	4	6.67
Delmarva Shelf	11	18.33

3.2 Jonesport

Of the 60 models run for the Jonesport site, 15 were well-modeled, and of those models, all used the Grossman and Ku equation. The Jonesport models were particularly successful when run with either temperature or both temperature and salinity/isotopes, and none of the salinity or isotope-only models were significant at this site. Additionally, models run under both annual and expert seasons were deemed well-modeled, and each of the four products had at least one well-modeled model at Jonesport, with the detailed breakdown of well-modeled pseudocarbonates by model input, equation, season, and product below (Table 6).

Table 6. *Well-Modeled Proxy System Models in Jonesport*

Data	Season	Method	Equation	r	p-value	RMSE	NRMSE
SODA	Annual	Temperature	GKM	0.457	0.004	0.614	1.440
SODA	Annual	Both	GKM	0.501	0.000	0.611	1.440
SODA	Expert	Temperature	GKM	0.346	0.025	0.632	1.430
SODA	Expert	Both	GKM	0.449	0.001	0.620	1.430
GLORYS	Annual	Temperature	GKM	0.391	0.048	0.844	1.320
GLORYS	Expert	Temperature	GKM	0.402	0.033	0.825	1.322
VIKING20X	Annual	Both	GKM	0.289	0.011	0.555	1.707
iCESM	Annual	Temperature(S)	GKM	0.345	0.011	0.953	1.830
iCESM	Annual	Both(S)	GKM	0.369	0.004	0.964	1.830
iCESM	Expert	Temperature(S)	GKM	0.275	0.035	0.949	1.841
iCESM	Expert	Both(S)	GKM	0.307	0.024	0.959	1.841
iCESM	Annual	Temperature(I)	GKM	0.345	0.011	1.220	1.830
iCESM	Annual	Both(I)	GKM	0.345	0.009	1.220	1.830
iCESM	Expert	Temperature(I)	GKM	0.275	0.044	1.216	1.841
iCESM	Expert	Both(I)	GKM	0.275	0.043	1.216	1.841

3.3 Isle au Haut

Of the five well-modeled pseudocarbonates at Isle au Haut site, all used VIKING20X for their data, and the majority of the significant models were based solely on salinity data. Though the product and model input results seemed to favor a specific group, both the annual and expert seasons were somewhat equally represented in the well-modeled pseudocarbonates, as were both the Grossman and Ku equation and the Williams et al. equation (Table 7).

Table 7. *Well-Modeled Proxy System Models in Isle au Haut*

Data	Season	Method	Equation	r	p-value	RMSE	NRMSE
VIKING20X	Annual	Salinity	GKM	0.304	0.033	0.462	1.834
VIKING20X	Expert	Salinity	GKM	0.351	0.012	0.461	1.833
VIKING20X	Annual	Salinity	WEA	0.304	0.023	0.442	0.465

Data	Season	Method	Equation	r	p-value	RMSE	NRMSE
VIKING20X	Expert	Salinity	WEA	0.351	0.013	0.441	0.469
VIKING20X	Expert	Both	WEA	0.286	0.047	0.459	0.469

3.4 Seguin

Ten of the 60 models run at the Seguin site were well-modeled, and of those ten, the majority of those models were run using the Grossman and Ku equation. VIKING20X produced the most well-modeled pseudocarbonates, while SODA and GLORYS also had well-modeled pseudocarbonates, though the reanalysis products were only significant with all model inputs present, while VIKING20X also produced well-modeled pseudocarbonates using only salinity. A similar difference can be seen between the equations, as the Williams et al. models were well-modeled with salinity only, while the Grossman and Ku models varied between needing only salinity and all model outputs. In similar fashion to the other sites, the annual and expert season both made an even appearance in the resulting well-modeled pseudocarbonates (Table 8).

Table 8. *Well-Modeled Proxy System Models in Seguin*

Data	Season	Method	Equation	r	p-value	RMSE	NRMSE
SODA	Annual	Both	GKM	0.473	0.031	0.394	1.930
SODA	Expert	Both	GKM	0.439	0.030	0.409	1.933
GLORYS	Annual	Both	GKM	0.620	0.005	0.315	1.720
GLORYS	Expert	Both	GKM	0.634	0.008	0.317	1.699
VIKING20X	Annual	Salinity	GKM	0.453	0.016	0.857	1.902
VIKING20X	Annual	Both	GKM	0.502	0.001	0.838	1.902
VIKING20X	Expert	Salinity	GKM	0.558	0.003	0.862	1.908
VIKING20X	Expert	Both	GKM	0.590	0.000	0.842	1.908
VIKING20X	Annual	Salinity	WEA	0.453	0.008	0.322	0.351
VIKING20X	Expert	Salinity	WEA	0.558	0.003	0.309	0.353

3.5 Georges Bank

Relatively few models in Georges Bank were deemed well-modeled, as only four models were significant. Of those four models, all used VIKING20X for their data, and all used the Williams et al. equation to calculate the pseudocarbonate. Once again, both the annual and expert season made equal appearances, however, the models were only well-modeled when temperature was present, whether by itself or in conjunction with salinity (Table 9).

Table 9. *Well-Modeled Proxy System Models in Georges Bank*

Data	Season	Method	Equation	r	p-value	RMSE	NRMSE
VIKING20X	Annual	Temperature	WEA	0.490	0.007	0.365	0.384
VIKING20X	Annual	Both	WEA	0.461	0.010	0.459	0.384
VIKING20X	Expert	Temperature	WEA	0.477	0.018	0.368	0.388
VIKING20X	Expert	Both	WEA	0.437	0.036	0.456	0.388

3.6 Long Island

Similar to Georges Bank, Long Island had a relatively low percentage of well-modeled pseudocarbonates, with only four models being significant. Of the four models, all used the Grossman and Ku equation, and all the models used data from the reanalysis products, whether that be SODA or GLORYS. The seasonality of the data did not seem to have an effect on how well-modeled the models were, since both annual and expert seasons are equally represented, but only the models with temperature as their sole input were deemed significant (Table 10).

Table 10. *Well-Modeled Proxy System Models in Long Island*

Data	Season	Method	Equation	r	p-value	RMSE	NRMSE
SODA	Annual	Temperature	GKM	0.493	0.002	0.502	2.201
SODA	Expert	Temperature	GKM	0.434	0.006	0.517	2.201
GLORYS	Annual	Temperature	GKM	0.592	0.006	0.429	2.091
GLORYS	Expert	Temperature	GKM	0.512	0.024	0.408	2.041

3.7 Delmarva Shelf

Of the 11 well-modeled pseudocarbonates at the Delmarva Shelf site, most used VIKING20X data while a handful used iCESM data. This was one of the only sites where iCESM data produced well-modeled pseudocarbonates, and the models built using iCESM depended on the presence of temperature, if not temperature coupled with salinity. For the VIKING20X models, every model input combination was represented, though when using the Grossman and Ku equation, only salinity was needed to make the model well-modeled, while when using Williams et al., any combination of temperature and salinity was significant. Additionally, all the iCESM models used the Grossman and Ku equation, while the seasonality once again did not seem to make a difference, as both are represented in the well-modeled pseudocarbonates relatively evenly (Table 11).

Table 11. *Well-Modeled Proxy System Models in Delmarva Shelf*

Data	Season	Method	Equation	r	p-value	RMSE	NRMSE
VIKING20X	Annual	Salinity	GKM	0.463	0.004	1.029	2.366
VIKING20X	Expert	Salinity	GKM	0.409	0.009	1.021	2.358
VIKING20X	Annual	Temperature	WEA	0.363	0.015	0.477	0.310
VIKING20X	Annual	Salinity	WEA	0.463	0.001	0.401	0.310
VIKING20X	Annual	Both	WEA	0.415	0.006	0.804	0.310
VIKING20X	Expert	Temperature	WEA	0.317	0.040	0.454	0.311
VIKING20X	Expert	Salinity	WEA	0.409	0.008	0.388	0.311
VIKING20X	Expert	Both	WEA	0.364	0.028	0.748	0.311
iCESM	Expert	Temperature(S)	GKM	0.185	0.046	1.782	2.575
iCESM	Expert	Temperature(I)	GKM	0.185	0.035	1.611	2.575
iCESM	Expert	Both(I)	GKM	0.185	0.039	1.611	2.575

4. Discussion

4.1 Model Input

Across the six sites, the effect of including either temperature, salinity, or isotope data on how well-modeled a certain pseudocarbonate was differed, and seemed to be linked to the products and equations used as well.

Temperature Driven

The oxygen isotope records from Georges Bank and Long Island seemed to be driven primarily by temperature variability, based on the statistically positive correlations between these records and the temperature-derived pseudocarbonate records. At both of these sites, the proxy system models were unable to produce a well-modeled pseudocarbonate using just salinity as an input. Temperature data needed to be included in the pseudocarbonate model, either by itself or in addition to the salinity data, in order to produce a well-modeled pseudocarbonate. Each of these sites yielded a relatively low number of well-modeled pseudocarbons, but did so under different conditions, with Georges Bank using exclusively VIKING20X with the Williams et al. equation and Long Island using only reanalysis data with the Grossman and Ku equation. However, at sites like Long Island, dating uncertainties might play into the low number of well-modeled pseudocarbons for the site-specific oxygen isotope records. These results suggest that at these two sites, variability in the isotope records is primarily caused by changes in temperature.

Salinity Driven

In contrast to the temperature driven oxygen isotope records of Georges Bank and Long Island, the oxygen isotope records from Isle au Haut and Seguin seemed to be driven primarily by salinity variability. In fact, at both sites, temperature data alone was unable to produce a well-modeled pseudocarbonate when put into the pseudocarbonate model, as salinity data was required to be a data input as well. The conditions under which the salinity-driven records correlated significantly with well-modeled pseudocarbons was more diverse than that of the temperature driven sites, as between the two sites, both VIKING20X and the two reanalysis products, SODA and GLORYS, were used. Additionally, both sites were able to use both the Williams et al. equation and the Grossman and Ku equation. This suggests that the models that depend on salinity are not as selective as the temperature driven models when using different equations, as temperature driven models needed a more bivalve-suited equation to produce well-modeled pseudocarbons.

Multi-factor Driven

The Jonesport and Delmarva Shelf oxygen isotope records both yielded the highest number of well-modeled pseudocarbons, and a much more diverse array of model inputs that were able to produce well-modeled pseudocarbons. Jonesport could be seen as more of a temperature driven record as well, since all well-modeled pseudocarbons there depended on at least the presence of temperature. However, we see that the Jonesport pseudocarbons were able to use both salinity data and isotope data to produce well-modeled pseudocarbons, a feature unique to these two multi-factor driven sites.

Delmarva Shelf, however, was the most diverse in terms of the well-modeled pseudocarbons that its record produced, as it was able to produce well-modeled pseudocarbons using only temperature, only salinity, or the combination of both, while using both salinity data and isotope data. This suggests that this site is influenced by both temperature and seawater oxygen isotope compositions, and so it represents changes in both of these hydrographic properties. No sites under any conditions were able to produce well-modeled pseudocarbons using just isotope data, since the variability between isotope values across each site's specific region was extremely low and not diverse enough to be different from the average metric taken over the entire time series.

4.2 Data Products

Reanalysis Products

In terms of the reanalysis products, SODA and GLORYS, we can see very similar pseudocarbonate modeling between the two products across the different well-modeled pseudocarbonates that they produced. In fact, both SODA and GLORYS performed at the same three sites, Seguin, Jonesport, and Long Island, and always appeared to produce well-modeled pseudocarbonates in the same places, which was not true for the climate models used, suggesting that the type of data product used might be more important than the type of product itself. Both products only worked in conjunction with the Grossman and Ku equation, and seemed to depend almost entirely on temperature, as none of the well-modeled pseudocarbonates using the reanalysis products had salinity as their only input. It is also worth noting that while these results might suggest that one pseudocarbonate model input (e.g., temperature or salinity) is more representative of a site's oxygen isotope record, it is also possible that the performance of a specific pseudocarbonate model input is a reflection of how well a data product performs in a certain area, not necessarily the model inputs that are used with the product.

Modeling Products

In terms of the modeling products, VIKING20X and iCESM/CESM1.2, pseudocarbonates created using data from these two products do not perform as similarly as those created using the reanalysis data, with VIKING20X being able to capture temperature and salinity variability more effectively than the iCESM. This is particularly due to the versatility of VIKING20X and the diversity of conditions under which it is able to explain variability in the oxygen isotope record. For example, VIKING20X performs well using both the Grossman and Ku equation and the Williams et al. equation, while the iCESM only works with the Grossman and Ku equation. In fact, VIKING20X is the only data product that is able to produce well-modeled pseudocarbonates with the Williams et al. equation.

Additionally, VIKING20X was able to perform at more sites than the iCESM, with the former producing well-modeled pseudocarbonates at five sites, nearly all, barring Georges Bank. The iCESM, however, was able to perform at Jonesport and Delmarva, the two multi-factor driven sites. Overall, VIKING20X was able to produce 24 well-modeled pseudocarbonates compared to the iCESM's 11, suggesting that VIKING20X might be more suitable to use in pseudocarbonate modeling in order to explain the variability of the shell oxygen isotope records. Additionally, the iCESM's poor performance may suggest that the model is unable to explain variability well for the oxygen isotope records. This disparity between the two modeling products may also be due to the very high spatial resolution of VIKING20X, which could contribute to its increased performance when compared with its lower resolution counterparts. In fact, higher resolution models have been shown to decrease circulation bias in the northwest Atlantic (Saba et al., 2015), and along with these results, suggest that coarser models are less suitable for proxy system modeling purposes.

4.3 Modeling Equations

The Grossman and Ku equation did a better job of producing well-modeled pseudocarbonates compared to the Williams equation, with 34 out of the 49 well-modeled pseudocarbonates using this equation. This is over twice as many as the Williams et al. equation, but when broken down by site, the two equations seem to follow an 'all-or-nothing' approach. For example, in the Jonesport and Long Island sites, every single well-modeled model uses Grossman and Ku, and this is also true at Seguin, barring two out of the ten models.

On the other hand, every single well-modeled model at Georges Bank uses the Williams et al. method. At Delmarva Shelf and Isle Au Haut we see more of a ‘mixed bag’ distribution between the two equations, but it suggests that either pseudocarbonate models at a specific site perform better when using one equation over the other one of the two equations, or that the choice of equation doesn’t necessarily have as much of an effect on how well-modeled the model was. Additionally, we can see that the sites that favor the Grossman and Ku equations tend to produce more well-modeled pseudocarbonates, suggesting that Grossman and Ku might be a better fit for producing well-modeled pseudocarbonates.

It’s also worth noting that the results suggest that the Grossman and Ku equation is better at capturing variability across the different data products. In fact, the Williams et al. equation produced well-modeled pseudocarbonates only when used in conjunction with VIKING20X, while the Grossman and Ku equation gave significant models with every data product used. This suggests that the Grossman and Ku equation better represents the relationship between water temperature, water isotopes, and shell oxygen isotopes.

This finding is consistent with the nature of the two equations used, as the Williams et al. equation has been historically used for corals (Evans et al., 2013; B. Williams, pers. comm., August 8, 2025), while the Grossman and Ku equation has been more popular for bivalves (Dettman et al, 1999; Walton, 2023). The results of this study support the notion that the Grossman and Ku equation is better suited to proxy system modeling with bivalve shells, and explains why the Grossman and Ku equation is quite robust when working with a more diverse range of climate data, like reanalysis data sets and lower-resolution products.

However, it is also important to note that while the variability in the oxygen isotope records at some sites was able to be modeled better under specific conditions (i.e., certain products or equations), these differences in variability might also be due to the oxygen isotope record itself, which may not be representing environmental conditions properly due to dating uncertainty and analytical errors. Therefore, though the results can suggest that certain model conditions are better at representing isotope record variability, further and more thorough study is needed to investigate the role of these model factors more and weigh their role with the oxygen isotope record itself.

4.4 Growing Season

One trend consistent across all sites, products, and equations was the relative neutrality of growing season. At every site, the proportion of well-modeled pseudocarbonates that used annual and expert dating was relatively if not exactly equal, which suggests that adjusting the timing of the pseudocarbonate calculation from January-December or October-September has relatively little impact on the accuracy of the pseudo carbonate itself. This is consistent with previous findings (Williams et al., 2024), which also shows significant models in both annual and expert seasons among bivalve pseudocarbonate in the Northern Atlantic.

4.5 Conclusions

As shown, a diverse array of combinations of different model inputs, products, and equations produced well-modeled pseudocarbonates at each of the six different sites. In particular, the results of this study suggest that the variability of the oxygen isotope records at certain sites are either temperature-driven, salinity-driven, or driven by a combination of both temperature and salinity, displaying that the records of certain sites rely on certain hydrographic properties to explain their variability well (i.e., produce a well-modeled pseudocarbonate). Additionally, the results of this study also suggest that the Grossman and Ku equation suited the bivalve pseudocarbonates more than the Williams et al. equation, displaying the increased modeling capability of the equation historically used for bivalves (Grossman and Ku) when compared to the equation historically used

for coral (Williams et al.). Though differences could be seen in the way that different products produced well-modeled pseudocarbonates with varying success, this variability did not extend to growing season, where annual season and expert season data performed largely the same.

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Author's Note

The GitHub for this project can be found at [this](#) link, and includes detailed code comments. Any additional questions about the project or code can be directed to sremaili@email.sc.edu.

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Appendix

Table 1A. *Western North Atlantic Proxy System Models*

Site	Data	Season	Method	Equation	r	p-value	RMSE	NRMSE
Delmarva Shelf	SODA	Annual	Temperature	GKM	0.037	0.828	0.563	2.263
Delmarva Shelf	SODA	Annual	Salinity	GKM	-0.037	0.818	0.517	2.263
Delmarva Shelf	SODA	Annual	Both	GKM	0.018	0.894	0.506	2.263
Delmarva Shelf	SODA	Expert	Temperature	GKM	0.050	0.766	0.559	2.262
Delmarva Shelf	SODA	Expert	Salinity	GKM	-0.087	0.609	0.537	2.262
Delmarva Shelf	SODA	Expert	Both	GKM	-0.009	0.938	0.514	2.262
Delmarva Shelf	SODA	Annual	Temperature	WEA	-0.037	0.828	0.474	0.349
Delmarva Shelf	SODA	Annual	Salinity	WEA	-0.037	0.809	0.426	0.349
Delmarva Shelf	SODA	Annual	Both	WEA	-0.039	0.828	0.632	0.349
Delmarva Shelf	SODA	Expert	Temperature	WEA	-0.050	0.786	0.483	0.351
Delmarva Shelf	SODA	Expert	Salinity	WEA	-0.087	0.631	0.425	0.351
Delmarva Shelf	SODA	Expert	Both	WEA	-0.068	0.687	0.639	0.351
Georges Bank	SODA	Annual	Temperature	GKM	-0.066	0.747	0.563	2.280
Georges Bank	SODA	Annual	Salinity	GKM	0.247	0.251	0.556	2.280
Georges Bank	SODA	Annual	Both	GKM	0.143	0.519	0.534	2.280
Georges Bank	SODA	Expert	Temperature	GKM	-0.050	0.854	0.510	2.253
Georges Bank	SODA	Expert	Salinity	GKM	0.081	0.699	0.563	2.253
Georges Bank	SODA	Expert	Both	GKM	0.002	0.947	0.520	2.253
Georges Bank	SODA	Annual	Temperature	WEA	0.066	0.750	0.397	0.344
Georges Bank	SODA	Annual	Salinity	WEA	0.247	0.253	0.333	0.344
Georges Bank	SODA	Annual	Both	WEA	0.149	0.484	0.467	0.344
Georges Bank	SODA	Expert	Temperature	WEA	0.050	0.836	0.377	0.339
Georges Bank	SODA	Expert	Salinity	WEA	0.081	0.718	0.338	0.339
Georges Bank	SODA	Expert	Both	WEA	0.067	0.765	0.442	0.339
Isle Au Haut	SODA	Annual	Temperature	GKM	-0.185	0.336	0.540	1.645
Isle Au Haut	SODA	Annual	Salinity	GKM	0.075	0.698	0.511	1.645
Isle Au Haut	SODA	Annual	Both	GKM	-0.097	0.628	0.542	1.645
Isle Au Haut	SODA	Expert	Temperature	GKM	-0.153	0.436	0.527	1.667
Isle Au Haut	SODA	Expert	Salinity	GKM	0.180	0.397	0.490	1.667
Isle Au Haut	SODA	Expert	Both	GKM	0.030	0.868	0.517	1.667
Isle Au Haut	SODA	Annual	Temperature	WEA	0.185	0.326	0.513	0.532
Isle Au Haut	SODA	Annual	Salinity	WEA	0.075	0.679	0.555	0.532
Isle Au Haut	SODA	Annual	Both	WEA	0.154	0.447	0.542	0.532
Isle Au Haut	SODA	Expert	Temperature	WEA	0.153	0.423	0.505	0.519
Isle Au Haut	SODA	Expert	Salinity	WEA	0.180	0.372	0.527	0.519
Isle Au Haut	SODA	Expert	Both	WEA	0.207	0.306	0.520	0.519
Jonesport	SODA	Annual	Temperature	GKM	0.457	0.004	0.614	1.440
Jonesport	SODA	Annual	Salinity	GKM	0.051	0.749	0.635	1.440
Jonesport	SODA	Annual	Both	GKM	0.501	0.000	0.611	1.440
Jonesport	SODA	Expert	Temperature	GKM	0.346	0.025	0.632	1.430
Jonesport	SODA	Expert	Salinity	GKM	0.151	0.378	0.620	1.430
Jonesport	SODA	Expert	Both	GKM	0.449	0.001	0.620	1.430
Jonesport	SODA	Annual	Temperature	WEA	-0.457	0.002	0.448	0.384
Jonesport	SODA	Annual	Salinity	WEA	0.051	0.732	0.396	0.384
Jonesport	SODA	Annual	Both	WEA	-0.268	0.098	0.472	0.384
Jonesport	SODA	Expert	Temperature	WEA	-0.346	0.030	0.435	0.388
Jonesport	SODA	Expert	Salinity	WEA	0.151	0.344	0.387	0.388
Jonesport	SODA	Expert	Both	WEA	-0.144	0.371	0.445	0.388
Long Island	SODA	Annual	Temperature	GKM	0.493	0.002	0.502	2.201
Long Island	SODA	Annual	Salinity	GKM	-0.462	0.003	0.645	2.201
Long Island	SODA	Annual	Both	GKM	0.113	0.512	0.567	2.201
Long Island	SODA	Expert	Temperature	GKM	0.434	0.006	0.517	2.201
Long Island	SODA	Expert	Salinity	GKM	-0.468	0.002	0.661	2.201

Long Island	SODA	Expert	Both	GKM	0.039	0.815	0.588	2.201
Long Island	SODA	Annual	Temperature	WEA	-0.493	0.004	0.602	0.462
Long Island	SODA	Annual	Salinity	WEA	-0.462	0.009	0.584	0.462
Long Island	SODA	Annual	Both	WEA	-0.554	0.000	0.733	0.462
Long Island	SODA	Expert	Temperature	WEA	-0.434	0.014	0.593	0.468
Long Island	SODA	Expert	Salinity	WEA	-0.468	0.006	0.588	0.468
Long Island	SODA	Expert	Both	WEA	-0.530	0.001	0.722	0.468
Seguin	SODA	Annual	Temperature	GKM	0.345	0.067	0.413	1.930
Seguin	SODA	Annual	Salinity	GKM	0.225	0.274	0.435	1.930
Seguin	SODA	Annual	Both	GKM	0.473	0.031	0.394	1.930
Seguin	SODA	Expert	Temperature	GKM	0.311	0.117	0.424	1.933
Seguin	SODA	Expert	Salinity	GKM	0.243	0.260	0.446	1.933
Seguin	SODA	Expert	Both	GKM	0.439	0.030	0.409	1.933
Seguin	SODA	Annual	Temperature	WEA	-0.345	0.076	0.480	0.409
Seguin	SODA	Annual	Salinity	WEA	0.225	0.280	0.403	0.409
Seguin	SODA	Annual	Both	WEA	-0.080	0.677	0.486	0.409
Seguin	SODA	Expert	Temperature	WEA	-0.311	0.130	0.480	0.414
Seguin	SODA	Expert	Salinity	WEA	0.243	0.246	0.407	0.414
Seguin	SODA	Expert	Both	WEA	-0.052	0.785	0.482	0.414
Delmarva Shelf	GLORYS	Annual	Temperature	GKM	-0.282	0.149	0.512	2.200
Delmarva Shelf	GLORYS	Annual	Salinity	GKM	0.202	0.302	0.433	2.200
Delmarva Shelf	GLORYS	Annual	Both	GKM	-0.260	0.172	0.428	2.200
Delmarva Shelf	GLORYS	Expert	Temperature	GKM	-0.278	0.149	0.498	2.202
Delmarva Shelf	GLORYS	Expert	Salinity	GKM	0.213	0.304	0.425	2.202
Delmarva Shelf	GLORYS	Expert	Both	GKM	-0.177	0.391	0.427	2.202
Delmarva Shelf	GLORYS	Annual	Temperature	WEA	0.282	0.164	0.439	0.395
Delmarva Shelf	GLORYS	Annual	Salinity	WEA	0.202	0.315	0.437	0.395
Delmarva Shelf	GLORYS	Annual	Both	WEA	0.251	0.196	0.589	0.395
Delmarva Shelf	GLORYS	Expert	Temperature	WEA	0.278	0.166	0.433	0.396
Delmarva Shelf	GLORYS	Expert	Salinity	WEA	0.213	0.316	0.437	0.396
Delmarva Shelf	GLORYS	Expert	Both	WEA	0.257	0.210	0.562	0.396
Georges Bank	GLORYS	Annual	Temperature	GKM	-0.303	0.181	0.543	2.159
Georges Bank	GLORYS	Annual	Salinity	GKM	0.278	0.217	0.565	2.159
Georges Bank	GLORYS	Annual	Both	GKM	-0.235	0.288	0.486	2.159
Georges Bank	GLORYS	Expert	Temperature	GKM	-0.313	0.192	0.534	2.160
Georges Bank	GLORYS	Expert	Salinity	GKM	0.237	0.294	0.571	2.160
Georges Bank	GLORYS	Expert	Both	GKM	-0.322	0.188	0.489	2.160
Georges Bank	GLORYS	Annual	Temperature	WEA	0.303	0.187	0.358	0.343
Georges Bank	GLORYS	Annual	Salinity	WEA	0.278	0.242	0.337	0.343
Georges Bank	GLORYS	Annual	Both	WEA	0.297	0.204	0.535	0.343
Georges Bank	GLORYS	Expert	Temperature	WEA	0.313	0.175	0.345	0.346
Georges Bank	GLORYS	Expert	Salinity	WEA	0.237	0.332	0.341	0.346
Georges Bank	GLORYS	Expert	Both	WEA	0.284	0.227	0.518	0.346
Isle Au Haut	GLORYS	Annual	Temperature	GKM	-0.202	0.519	0.629	1.425
Isle Au Haut	GLORYS	Annual	Salinity	GKM	-0.022	0.920	0.516	1.425
Isle Au Haut	GLORYS	Annual	Both	GKM	-0.183	0.580	0.598	1.425
Isle Au Haut	GLORYS	Expert	Temperature	GKM	-0.517	0.094	0.656	1.367
Isle Au Haut	GLORYS	Expert	Salinity	GKM	-0.044	0.887	0.524	1.367
Isle Au Haut	GLORYS	Expert	Both	GKM	-0.359	0.264	0.633	1.367
Isle Au Haut	GLORYS	Annual	Temperature	WEA	0.202	0.495	0.466	0.538
Isle Au Haut	GLORYS	Annual	Salinity	WEA	-0.022	0.929	0.528	0.538
Isle Au Haut	GLORYS	Annual	Both	WEA	0.087	0.788	0.486	0.538
Isle Au Haut	GLORYS	Expert	Temperature	WEA	0.517	0.092	0.454	0.556
Isle Au Haut	GLORYS	Expert	Salinity	WEA	-0.044	0.860	0.546	0.556
Isle Au Haut	GLORYS	Expert	Both	WEA	0.204	0.526	0.465	0.556
Jonesport	GLORYS	Annual	Temperature	GKM	0.391	0.048	0.844	1.320
Jonesport	GLORYS	Annual	Salinity	GKM	-0.416	0.039	0.843	1.320
Jonesport	GLORYS	Annual	Both	GKM	0.178	0.358	0.840	1.320

Jonesport	GLORYS	Expert	Temperature	GKM	0.402	0.033	0.825	1.322
Jonesport	GLORYS	Expert	Salinity	GKM	-0.256	0.199	0.844	1.322
Jonesport	GLORYS	Expert	Both	GKM	0.246	0.212	0.831	1.322
Jonesport	GLORYS	Annual	Temperature	WEA	-0.391	0.050	0.485	0.433
Jonesport	GLORYS	Annual	Salinity	WEA	-0.416	0.033	0.465	0.433
Jonesport	GLORYS	Annual	Both	WEA	-0.439	0.021	0.537	0.433
Jonesport	GLORYS	Expert	Temperature	WEA	-0.402	0.035	0.484	0.435
Jonesport	GLORYS	Expert	Salinity	WEA	-0.256	0.202	0.459	0.435
Jonesport	GLORYS	Expert	Both	WEA	-0.391	0.042	0.521	0.435
Long Island	GLORYS	Annual	Temperature	GKM	0.592	0.006	0.429	2.091
Long Island	GLORYS	Annual	Salinity	GKM	-0.584	0.003	0.656	2.091
Long Island	GLORYS	Annual	Both	GKM	0.062	0.791	0.528	2.091
Long Island	GLORYS	Expert	Temperature	GKM	0.512	0.024	0.408	2.041
Long Island	GLORYS	Expert	Salinity	GKM	-0.564	0.009	0.596	2.041
Long Island	GLORYS	Expert	Both	GKM	-0.019	0.913	0.496	2.041
Long Island	GLORYS	Annual	Temperature	WEA	-0.592	0.003	0.614	0.446
Long Island	GLORYS	Annual	Salinity	WEA	-0.584	0.004	0.616	0.446
Long Island	GLORYS	Annual	Both	WEA	-0.641	0.001	0.795	0.446
Long Island	GLORYS	Expert	Temperature	WEA	-0.512	0.021	0.550	0.401
Long Island	GLORYS	Expert	Salinity	WEA	-0.564	0.010	0.565	0.401
Long Island	GLORYS	Expert	Both	WEA	-0.601	0.002	0.723	0.401
Seguin	GLORYS	Annual	Temperature	GKM	0.338	0.165	0.367	1.720
Seguin	GLORYS	Annual	Salinity	GKM	0.285	0.243	0.345	1.720
Seguin	GLORYS	Annual	Both	GKM	0.620	0.005	0.315	1.720
Seguin	GLORYS	Expert	Temperature	GKM	0.332	0.196	0.368	1.699
Seguin	GLORYS	Expert	Salinity	GKM	0.377	0.121	0.333	1.699
Seguin	GLORYS	Expert	Both	GKM	0.634	0.008	0.317	1.699
Seguin	GLORYS	Annual	Temperature	WEA	-0.338	0.127	0.387	0.303
Seguin	GLORYS	Annual	Salinity	WEA	0.285	0.233	0.294	0.303
Seguin	GLORYS	Annual	Both	WEA	-0.055	0.792	0.410	0.303
Seguin	GLORYS	Expert	Temperature	WEA	-0.332	0.189	0.367	0.299
Seguin	GLORYS	Expert	Salinity	WEA	0.377	0.116	0.276	0.299
Seguin	GLORYS	Expert	Both	WEA	0.026	0.908	0.370	0.299
Delmarva Shelf	VIKING20X	Annual	Temperature	GKM	-0.363	0.009	1.157	2.366
Delmarva Shelf	VIKING20X	Annual	Salinity	GKM	0.463	0.004	1.029	2.366
Delmarva Shelf	VIKING20X	Annual	Both	GKM	-0.021	0.885	1.021	2.366
Delmarva Shelf	VIKING20X	Expert	Temperature	GKM	-0.317	0.049	1.125	2.358
Delmarva Shelf	VIKING20X	Expert	Salinity	GKM	0.409	0.009	1.021	2.358
Delmarva Shelf	VIKING20X	Expert	Both	GKM	0.000	0.960	1.010	2.358
Delmarva Shelf	VIKING20X	Annual	Temperature	WEA	0.363	0.015	0.477	0.310
Delmarva Shelf	VIKING20X	Annual	Salinity	WEA	0.463	0.001	0.401	0.310
Delmarva Shelf	VIKING20X	Annual	Both	WEA	0.415	0.006	0.804	0.310
Delmarva Shelf	VIKING20X	Expert	Temperature	WEA	0.317	0.040	0.454	0.311
Delmarva Shelf	VIKING20X	Expert	Salinity	WEA	0.409	0.008	0.388	0.311
Delmarva Shelf	VIKING20X	Expert	Both	WEA	0.364	0.028	0.748	0.311
Georges Bank	VIKING20X	Annual	Temperature	GKM	-0.490	0.003	0.831	2.327
Georges Bank	VIKING20X	Annual	Salinity	GKM	0.357	0.052	0.675	2.327
Georges Bank	VIKING20X	Annual	Both	GKM	-0.488	0.013	0.764	2.327
Georges Bank	VIKING20X	Expert	Temperature	GKM	-0.477	0.011	0.827	2.326
Georges Bank	VIKING20X	Expert	Salinity	GKM	0.316	0.120	0.683	2.326
Georges Bank	VIKING20X	Expert	Both	GKM	-0.522	0.006	0.769	2.326
Georges Bank	VIKING20X	Annual	Temperature	WEA	0.490	0.007	0.365	0.384
Georges Bank	VIKING20X	Annual	Salinity	WEA	0.357	0.059	0.363	0.384
Georges Bank	VIKING20X	Annual	Both	WEA	0.461	0.010	0.459	0.384
Georges Bank	VIKING20X	Expert	Temperature	WEA	0.477	0.018	0.368	0.388
Georges Bank	VIKING20X	Expert	Salinity	WEA	0.316	0.118	0.368	0.388
Georges Bank	VIKING20X	Expert	Both	WEA	0.437	0.036	0.456	0.388
Isle Au Haut	VIKING20X	Annual	Temperature	GKM	-0.112	0.439	0.507	1.834

Isle Au Haut	VIKING20X	Annual	Salinity	GKM	0.304	0.033	0.462	1.834
Isle Au Haut	VIKING20X	Annual	Both	GKM	0.167	0.251	0.472	1.834
Isle Au Haut	VIKING20X	Expert	Temperature	GKM	-0.116	0.446	0.511	1.833
Isle Au Haut	VIKING20X	Expert	Salinity	GKM	0.351	0.012	0.461	1.833
Isle Au Haut	VIKING20X	Expert	Both	GKM	0.197	0.184	0.472	1.833
Isle Au Haut	VIKING20X	Annual	Temperature	WEA	0.112	0.426	0.472	0.465
Isle Au Haut	VIKING20X	Annual	Salinity	WEA	0.304	0.023	0.442	0.465
Isle Au Haut	VIKING20X	Annual	Both	WEA	0.258	0.057	0.463	0.465
Isle Au Haut	VIKING20X	Expert	Temperature	WEA	0.116	0.418	0.475	0.469
Isle Au Haut	VIKING20X	Expert	Salinity	WEA	0.351	0.013	0.441	0.469
Isle Au Haut	VIKING20X	Expert	Both	WEA	0.286	0.047	0.459	0.469
Jonesport	VIKING20X	Annual	Temperature	GKM	0.176	0.147	0.575	1.707
Jonesport	VIKING20X	Annual	Salinity	GKM	0.047	0.701	0.578	1.707
Jonesport	VIKING20X	Annual	Both	GKM	0.289	0.011	0.555	1.707
Jonesport	VIKING20X	Expert	Temperature	GKM	0.151	0.212	0.580	1.699
Jonesport	VIKING20X	Expert	Salinity	GKM	0.029	0.803	0.581	1.699
Jonesport	VIKING20X	Expert	Both	GKM	0.242	0.050	0.562	1.699
Jonesport	VIKING20X	Annual	Temperature	WEA	-0.176	0.141	0.504	0.439
Jonesport	VIKING20X	Annual	Salinity	WEA	0.047	0.707	0.452	0.439
Jonesport	VIKING20X	Annual	Both	WEA	-0.090	0.447	0.547	0.439
Jonesport	VIKING20X	Expert	Temperature	WEA	-0.151	0.199	0.497	0.438
Jonesport	VIKING20X	Expert	Salinity	WEA	0.029	0.808	0.453	0.438
Jonesport	VIKING20X	Expert	Both	WEA	-0.082	0.515	0.540	0.438
Long Island	VIKING20X	Annual	Temperature	GKM	0.154	0.287	1.188	2.104
Long Island	VIKING20X	Annual	Salinity	GKM	-0.108	0.517	1.214	2.104
Long Island	VIKING20X	Annual	Both	GKM	0.056	0.698	1.190	2.104
Long Island	VIKING20X	Expert	Temperature	GKM	0.110	0.449	1.204	2.114
Long Island	VIKING20X	Expert	Salinity	GKM	-0.017	0.902	1.222	2.114
Long Island	VIKING20X	Expert	Both	GKM	0.104	0.451	1.202	2.114
Long Island	VIKING20X	Annual	Temperature	WEA	-0.154	0.258	0.488	0.413
Long Island	VIKING20X	Annual	Salinity	WEA	-0.108	0.514	0.490	0.413
Long Island	VIKING20X	Annual	Both	WEA	-0.144	0.340	0.604	0.413
Long Island	VIKING20X	Expert	Temperature	WEA	-0.110	0.458	0.471	0.410
Long Island	VIKING20X	Expert	Salinity	WEA	-0.017	0.902	0.468	0.410
Long Island	VIKING20X	Expert	Both	WEA	-0.067	0.691	0.568	0.410
Seguin	VIKING20X	Annual	Temperature	GKM	0.240	0.156	0.848	1.902
Seguin	VIKING20X	Annual	Salinity	GKM	0.453	0.016	0.857	1.902
Seguin	VIKING20X	Annual	Both	GKM	0.502	0.001	0.838	1.902
Seguin	VIKING20X	Expert	Temperature	GKM	0.260	0.142	0.858	1.908
Seguin	VIKING20X	Expert	Salinity	GKM	0.558	0.003	0.862	1.908
Seguin	VIKING20X	Expert	Both	GKM	0.590	0.000	0.842	1.908
Seguin	VIKING20X	Annual	Temperature	WEA	-0.240	0.143	0.406	0.351
Seguin	VIKING20X	Annual	Salinity	WEA	0.453	0.008	0.322	0.351
Seguin	VIKING20X	Annual	Both	WEA	0.226	0.243	0.382	0.351
Seguin	VIKING20X	Expert	Temperature	WEA	-0.260	0.127	0.410	0.353
Seguin	VIKING20X	Expert	Salinity	WEA	0.558	0.003	0.309	0.353
Seguin	VIKING20X	Expert	Both	WEA	0.295	0.139	0.372	0.353
Delmarva Shelf	iCESM	Annual	Temperature(S)	GKM	0.085	0.359	1.781	2.574
Delmarva Shelf	iCESM	Annual	Salinity	GKM	-0.384	0.001	1.785	2.574
Delmarva Shelf	iCESM	Annual	Both(S)	GKM	-0.052	0.561	1.783	2.574
Delmarva Shelf	iCESM	Expert	Temperature(S)	GKM	0.185	0.046	1.782	2.575
Delmarva Shelf	iCESM	Expert	Salinity	GKM	-0.382	0.000	1.786	2.575
Delmarva Shelf	iCESM	Expert	Both(S)	GKM	0.022	0.800	1.785	2.575
Delmarva Shelf	iCESM	Annual	Temperature(I)	GKM	0.085	0.364	1.610	2.574
Delmarva Shelf	iCESM	Annual	Isotopes	GKM	-0.041	0.672	1.612	2.574
Delmarva Shelf	iCESM	Annual	Both(I)	GKM	0.085	0.357	1.610	2.574
Delmarva Shelf	iCESM	Expert	Temperature(I)	GKM	0.185	0.035	1.611	2.575
Delmarva Shelf	iCESM	Expert	Isotopes	GKM	-0.006	0.926	1.612	2.575

Delmarva Shelf	iCESM	Expert	Both(I)	GKM	0.185	0.039	1.611	2.575
Delmarva Shelf	iCESM	Annual	Temperature(S)	WEA	-0.169	0.064	0.381	0.358
Delmarva Shelf	iCESM	Annual	Salinity	WEA	-0.381	0.000	0.375	0.358
Delmarva Shelf	iCESM	Annual	Both(S)	WEA	-0.289	0.003	0.399	0.358
Delmarva Shelf	iCESM	Expert	Temperature(S)	WEA	-0.185	0.037	0.382	0.358
Delmarva Shelf	iCESM	Expert	Salinity	WEA	-0.382	0.000	0.374	0.358
Delmarva Shelf	iCESM	Expert	Both(S)	WEA	-0.307	0.000	0.399	0.358
Delmarva Shelf	iCESM	Annual	Temperature(I)	WEA	-0.169	0.058	0.381	0.358
Delmarva Shelf	iCESM	Annual	Isotopes	WEA	-0.032	0.756	0.358	0.358
Delmarva Shelf	iCESM	Annual	Both(I)	WEA	-0.169	0.056	0.381	0.358
Delmarva Shelf	iCESM	Expert	Temperature(I)	WEA	-0.185	0.027	0.382	0.358
Delmarva Shelf	iCESM	Expert	Isotopes	WEA	-0.006	0.923	0.358	0.358
Delmarva Shelf	iCESM	Expert	Both(I)	WEA	-0.185	0.035	0.382	0.358
Georges Bank	iCESM	Annual	Temperature(S)	GKM	0.141	0.315	0.376	2.367
Georges Bank	iCESM	Annual	Salinity	GKM	-0.033	0.825	0.381	2.367
Georges Bank	iCESM	Annual	Both(S)	GKM	0.173	0.247	0.374	2.367
Georges Bank	iCESM	Expert	Temperature(S)	GKM	0.131	0.407	0.383	2.361
Georges Bank	iCESM	Expert	Salinity	GKM	-0.019	0.892	0.388	2.361
Georges Bank	iCESM	Expert	Both(S)	GKM	0.166	0.276	0.381	2.361
Georges Bank	iCESM	Annual	Temperature(I)	GKM	0.141	0.341	0.673	2.367
Georges Bank	iCESM	Annual	Isotopes	GKM	-0.118	0.468	0.674	2.367
Georges Bank	iCESM	Annual	Both(I)	GKM	0.141	0.348	0.673	2.367
Georges Bank	iCESM	Expert	Temperature(I)	GKM	0.131	0.386	0.680	2.361
Georges Bank	iCESM	Expert	Isotopes	GKM	-0.095	0.585	0.682	2.361
Georges Bank	iCESM	Expert	Both(I)	GKM	0.131	0.407	0.680	2.361
Georges Bank	iCESM	Annual	Temperature(S)	WEA	-0.131	0.401	0.387	0.369
Georges Bank	iCESM	Annual	Salinity	WEA	-0.024	0.879	0.372	0.369
Georges Bank	iCESM	Annual	Both(S)	WEA	-0.104	0.512	0.395	0.369
Georges Bank	iCESM	Expert	Temperature(S)	WEA	-0.131	0.391	0.393	0.375
Georges Bank	iCESM	Expert	Salinity	WEA	-0.019	0.886	0.378	0.375
Georges Bank	iCESM	Expert	Both(S)	WEA	-0.103	0.510	0.400	0.375
Georges Bank	iCESM	Annual	Temperature(I)	WEA	-0.131	0.382	0.387	0.369
Georges Bank	iCESM	Annual	Isotopes	WEA	-0.114	0.467	0.369	0.369
Georges Bank	iCESM	Annual	Both(I)	WEA	-0.131	0.369	0.387	0.369
Georges Bank	iCESM	Expert	Temperature(I)	WEA	-0.131	0.420	0.393	0.375
Georges Bank	iCESM	Expert	Isotopes	WEA	-0.095	0.573	0.375	0.375
Georges Bank	iCESM	Expert	Both(I)	WEA	-0.131	0.376	0.393	0.375
Isle Au Haut	iCESM	Annual	Temperature(S)	GKM	0.200	0.111	0.542	1.790
Isle Au Haut	iCESM	Annual	Salinity	GKM	-0.159	0.218	0.562	1.790
Isle Au Haut	iCESM	Annual	Both(S)	GKM	0.134	0.267	0.553	1.790
Isle Au Haut	iCESM	Expert	Temperature(S)	GKM	0.169	0.187	0.544	1.790
Isle Au Haut	iCESM	Expert	Salinity	GKM	-0.153	0.224	0.562	1.790
Isle Au Haut	iCESM	Expert	Both(S)	GKM	0.102	0.409	0.554	1.790
Isle Au Haut	iCESM	Annual	Temperature(I)	GKM	0.200	0.110	0.771	1.790
Isle Au Haut	iCESM	Annual	Isotopes	GKM	0.084	0.488	0.778	1.790
Isle Au Haut	iCESM	Annual	Both(I)	GKM	0.200	0.111	0.771	1.790
Isle Au Haut	iCESM	Expert	Temperature(I)	GKM	0.169	0.198	0.771	1.790
Isle Au Haut	iCESM	Expert	Isotopes	GKM	0.149	0.202	0.778	1.790
Isle Au Haut	iCESM	Expert	Both(I)	GKM	0.169	0.176	0.771	1.790
Isle Au Haut	iCESM	Annual	Temperature(S)	WEA	-0.200	0.127	0.417	0.393
Isle Au Haut	iCESM	Annual	Salinity	WEA	-0.159	0.225	0.401	0.393
Isle Au Haut	iCESM	Annual	Both(S)	WEA	-0.229	0.085	0.427	0.393
Isle Au Haut	iCESM	Expert	Temperature(S)	WEA	-0.169	0.173	0.415	0.393
Isle Au Haut	iCESM	Expert	Salinity	WEA	-0.153	0.249	0.401	0.393
Isle Au Haut	iCESM	Expert	Both(S)	WEA	-0.201	0.112	0.424	0.393
Isle Au Haut	iCESM	Annual	Temperature(I)	WEA	-0.200	0.104	0.417	0.393
Isle Au Haut	iCESM	Annual	Isotopes	WEA	0.084	0.503	0.393	0.393
Isle Au Haut	iCESM	Annual	Both(I)	WEA	-0.200	0.116	0.417	0.393

Isle Au Haut	iCESM	Expert	Temperature(I)	WEA	-0.169	0.176	0.415	0.393
Isle Au Haut	iCESM	Expert	Isotopes	WEA	0.150	0.214	0.393	0.393
Isle Au Haut	iCESM	Expert	Both(I)	WEA	-0.169	0.160	0.415	0.393
Jonesport	iCESM	Annual	Temperature(S)	GKM	0.345	0.011	0.953	1.830
Jonesport	iCESM	Annual	Salinity	GKM	0.017	0.897	0.976	1.830
Jonesport	iCESM	Annual	Both(S)	GKM	0.369	0.004	0.964	1.830
Jonesport	iCESM	Expert	Temperature(S)	GKM	0.275	0.035	0.949	1.841
Jonesport	iCESM	Expert	Salinity	GKM	0.066	0.633	0.960	1.841
Jonesport	iCESM	Expert	Both(S)	GKM	0.307	0.024	0.959	1.841
Jonesport	iCESM	Annual	Temperature(I)	GKM	0.345	0.011	1.220	1.830
Jonesport	iCESM	Annual	Isotopes	GKM	-0.120	0.402	1.229	1.830
Jonesport	iCESM	Annual	Both(I)	GKM	0.345	0.009	1.220	1.830
Jonesport	iCESM	Expert	Temperature(I)	GKM	0.275	0.044	1.216	1.841
Jonesport	iCESM	Expert	Isotopes	GKM	-0.136	0.330	1.215	1.841
Jonesport	iCESM	Expert	Both(I)	GKM	0.275	0.043	1.216	1.841
Jonesport	iCESM	Annual	Temperature(S)	WEA	-0.278	0.035	0.470	0.429
Jonesport	iCESM	Annual	Salinity	WEA	0.035	0.787	0.426	0.429
Jonesport	iCESM	Annual	Both(S)	WEA	-0.232	0.083	0.469	0.429
Jonesport	iCESM	Expert	Temperature(S)	WEA	-0.275	0.041	0.465	0.427
Jonesport	iCESM	Expert	Salinity	WEA	0.066	0.621	0.423	0.427
Jonesport	iCESM	Expert	Both(S)	WEA	-0.216	0.096	0.463	0.427
Jonesport	iCESM	Annual	Temperature(I)	WEA	-0.278	0.030	0.470	0.429
Jonesport	iCESM	Annual	Isotopes	WEA	-0.103	0.448	0.429	0.429
Jonesport	iCESM	Annual	Both(I)	WEA	-0.278	0.036	0.470	0.429
Jonesport	iCESM	Expert	Temperature(I)	WEA	-0.275	0.048	0.465	0.427
Jonesport	iCESM	Expert	Isotopes	WEA	-0.137	0.342	0.427	0.427
Jonesport	iCESM	Expert	Both(I)	WEA	-0.275	0.041	0.465	0.427
Long Island	iCESM	Annual	Temperature(S)	GKM	0.023	0.842	1.735	2.017
Long Island	iCESM	Annual	Salinity	GKM	-0.004	0.942	1.696	2.017
Long Island	iCESM	Annual	Both(S)	GKM	0.026	0.827	1.730	2.017
Long Island	iCESM	Expert	Temperature(S)	GKM	0.031	0.795	1.733	2.019
Long Island	iCESM	Expert	Salinity	GKM	-0.015	0.904	1.696	2.019
Long Island	iCESM	Expert	Both(S)	GKM	0.034	0.787	1.729	2.019
Long Island	iCESM	Annual	Temperature(I)	GKM	0.023	0.848	2.450	2.017
Long Island	iCESM	Annual	Isotopes	GKM	-0.121	0.314	2.414	2.017
Long Island	iCESM	Annual	Both(I)	GKM	0.022	0.842	2.450	2.017
Long Island	iCESM	Expert	Temperature(I)	GKM	0.031	0.789	2.448	2.019
Long Island	iCESM	Expert	Isotopes	GKM	-0.142	0.217	2.414	2.019
Long Island	iCESM	Expert	Both(I)	GKM	0.031	0.783	2.448	2.019
Long Island	iCESM	Annual	Temperature(S)	WEA	-0.025	0.825	0.537	0.522
Long Island	iCESM	Annual	Salinity	WEA	-0.006	0.936	0.522	0.522
Long Island	iCESM	Annual	Both(S)	WEA	-0.022	0.836	0.541	0.522
Long Island	iCESM	Expert	Temperature(S)	WEA	-0.031	0.818	0.539	0.524
Long Island	iCESM	Expert	Salinity	WEA	-0.015	0.884	0.525	0.524
Long Island	iCESM	Expert	Both(S)	WEA	-0.029	0.789	0.543	0.524
Long Island	iCESM	Annual	Temperature(I)	WEA	-0.025	0.837	0.537	0.522
Long Island	iCESM	Annual	Isotopes	WEA	-0.122	0.313	0.522	0.522
Long Island	iCESM	Annual	Both(I)	WEA	-0.025	0.838	0.537	0.522
Long Island	iCESM	Expert	Temperature(I)	WEA	-0.031	0.816	0.539	0.524
Long Island	iCESM	Expert	Isotopes	WEA	-0.142	0.238	0.524	0.524
Long Island	iCESM	Expert	Both(I)	WEA	-0.031	0.796	0.539	0.524
Seguin	iCESM	Annual	Temperature(S)	GKM	0.160	0.087	0.664	1.991
Seguin	iCESM	Annual	Salinity	GKM	-0.146	0.149	0.668	1.991
Seguin	iCESM	Annual	Both(S)	GKM	0.116	0.180	0.666	1.991
Seguin	iCESM	Expert	Temperature(S)	GKM	0.165	0.107	0.659	1.996
Seguin	iCESM	Expert	Salinity	GKM	-0.132	0.187	0.661	1.996
Seguin	iCESM	Expert	Both(S)	GKM	0.125	0.149	0.660	1.996
Seguin	iCESM	Annual	Temperature(I)	GKM	0.160	0.073	0.924	1.991

Seguin	iCESM	Annual	Isotopes	GKM	-0.034	0.727	0.925	1.991
Seguin	iCESM	Annual	Both(I)	GKM	0.160	0.073	0.924	1.991
Seguin	iCESM	Expert	Temperature(I)	GKM	0.165	0.084	0.919	1.996
Seguin	iCESM	Expert	Isotopes	GKM	-0.034	0.710	0.919	1.996
Seguin	iCESM	Expert	Both(I)	GKM	0.165	0.077	0.919	1.996
Seguin	iCESM	Annual	Temperature(S)	WEA	-0.149	0.091	0.395	0.373
Seguin	iCESM	Annual	Salinity	WEA	-0.144	0.146	0.378	0.373
Seguin	iCESM	Annual	Both(S)	WEA	-0.174	0.065	0.403	0.373
Seguin	iCESM	Expert	Temperature(S)	WEA	-0.165	0.066	0.396	0.374
Seguin	iCESM	Expert	Salinity	WEA	-0.132	0.208	0.380	0.374
Seguin	iCESM	Expert	Both(S)	WEA	-0.182	0.051	0.404	0.374
Seguin	iCESM	Annual	Temperature(I)	WEA	-0.149	0.095	0.395	0.373
Seguin	iCESM	Annual	Isotopes	WEA	-0.031	0.743	0.373	0.373
Seguin	iCESM	Annual	Both(I)	WEA	-0.149	0.091	0.395	0.373
Seguin	iCESM	Expert	Temperature(I)	WEA	-0.165	0.076	0.396	0.374
Seguin	iCESM	Expert	Isotopes	WEA	-0.034	0.710	0.374	0.374
Seguin	iCESM	Expert	Both(I)	WEA	-0.165	0.089	0.396	0.374
